

Project ID:

1. Topic (12 words max)

AI-Powered Personalized Course Creation and Discovery Platform with Adaptive Learning Paths

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	R26-SE-041
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Higher education institutions globally face persistent challenges in student retention, engagement, and academic progression, exacerbated by diverse student populations with varying learning needs, paces, and backgrounds. The traditional "one-size-fits-all" pedagogical model struggles to address this diversity, often leading to disengagement, knowledge gaps, and suboptimal learning outcomes. While technology-enhanced learning and AI present transformative opportunities, current solutions are fragmented. Many platforms focus either on simple content delivery, static tutoring, or automated assessment without integrating **true personalization, dynamic pathway adaptation, and intelligent content discovery** into a cohesive ecosystem.

The core problem is the absence of an intelligent platform that **simultaneously empowers students to create personalized adaptive learning paths**. Existing tools often address only one side of this equation—either content creation or learner tutoring—leading to a disconnect between instructional design and the adaptive learning experience. This gap prevents the realization of a fully responsive educational environment where learning content, pace, and sequence are dynamically tailored in real-time based on continuous analysis of individual learner performance, confidence, and behavior.

Nature/Gravity of the Problem:

The primary users affected are **university students** (especially undergraduates) and **educators**. Students from non-traditional backgrounds or with different learning styles are disproportionately impacted by rigid curricula. A platform serving these user personas must address student needs for guided, self-paced discovery and educator needs for intuitive, AI-assisted course authoring.

Impact of the Problem:

- **Social & Educational:** Contributes to equity issues in education, as uniform teaching can widen the achievement gap. It also fails to foster crucial self-directed learning skills and deep engagement.
- **Technical:** Highlights a gap in integrating advanced AI (like reinforcement learning for path optimization and NLP for affective sensing) with proven pedagogical frameworks and learning analytics.
- **Economic:** Inefficient learning leads to longer time-to-competency, affecting graduate readiness for the workforce. For institutions, poor student progression impacts funding and reputation.
- **Ethical:** Raises concerns about data privacy, algorithmic bias in personalized recommendations, and the risk of student over-reliance on AI, potentially diminishing critical thinking.

FoC Research Cluster and Industry Vertical:

This problem aligns perfectly with the **Centre of Excellence in AI (CEAI)**. It requires research in core AI domains like machine learning (for adaptive algorithms), natural language processing (for content understanding and interaction), and data analytics (for learning analytics and insight generation). The primary industry vertical is **EdTech**, specifically targeting **Higher Education and Lifelong Learning**.

Contribution to Sustainable Development Goals (SDGs):

This project directly advances **SDG 4 (Quality Education)** by aiming to make inclusive, equitable, and effective learning a reality through personalized adaptation. It promotes lifelong learning opportunities (SDG 4.4) and leverages technology to enhance educational delivery (SDG 4.c).

References.

IEEE References:

1. W. Cui, Z. Xue and K. -P. Thai, "Performance Comparison of an AI-Based Adaptive Learning System in China," *2018 Chinese Automation Congress (CAC)*, Xi'an, China, 2018, pp. 3170-3175, doi: 10.1109/CAC.2018.8623327.
2. H. Fang, J. Liu, R. Huang and Y. Li, "Research on adaptive mobile learning resources platform based on learning object model," *2009 International Conference on Application of Information and Communication Technologies*, Baku, Azerbaijan, 2009, pp. 1-6, doi: 10.1109/ICAICT.2009.5372533.
3. M. Y. Al Balushi and A. S. Al Harthi, "Adaptive Learning Pathways: Advancing Personalization in Online Education Platforms," *2024 2nd International Conference on Computing and Data Analytics (ICCD A)*, Shinas, Oman, 2024, pp. 1-6, doi: 10.1109/ICCD A64887.2024.10867273.
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5. T. M. N. Marouf *et al.*, "Mobile and e-Learning Platforms Utilizing Adaptive Technologies to Personalize Education in Remote Areas," *2025 3rd International Conference on Business Analytics for Technology and Security (ICBATS)*, Dubai, United Arab Emirates, 2025, pp. 1-8, doi: 10.1109/ICBATS66542.2025.11258392.
6. M. H. Sallaah, V. Akila, P. Sharma, S. S. Kumar and K. R. Singh, "Deep Reinforcement Learning for Personalized AI Tutors in Intelligent Education Systems and Adaptive E-Learning Platforms," *2025 3rd International Conference on Cyber Resilience (ICCR)*, Dubai, United Arab Emirates, 2025, pp. 1-8, doi: 10.1109/ICCR67387.2025.11291670.
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- 8.** R. Majumdar, K. Takami and H. Ogata, "Learning with Explainable AI-Recommendations at School: Extracting Patterns of Self-Directed Learning from Learning Logs," *2023 IEEE International Conference on Advanced Learning Technologies (ICALT)*, Orem, UT, USA, 2023, pp. 245-249, doi: 10.1109/ICALT58122.2023.00078.
- 9.** G. Shen, S. Wang and X. Gan, "Development and Application of Intelligent Tutoring System Based on AI-assisted Robots," *2025 7th International Conference on Computer Science and Technologies in Education (CSTE)*, Wuhan, China, 2025, pp. 385-390, doi: 10.1109/CSTE64638.2025.11091958.
- 10.** T. H. Teo and Z. J. Chow, "Student Perspectives on AI-Powered Learning Tools: An Exploratory Study of Deepseek, ChatGPT, and Gemini in Educational Settings," *2025 IEEE 18th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc)*, Singapore, Singapore, 2025, pp. 98-104, doi: 10.1109/MCSoc67473.2025.00026.
- 11.** B. Rozputnia, L. Shevchenko, V. Umanets, Y. Sabadosh, A. Kushnir and N. Voitsekhivska, "Future Trends of AI in Higher Education: Google Workspace and Gemini," *2025 15th International Conference on Advanced Computer Information Technologies (ACIT)*, Sibenik, Croatia, 2025, pp. 1019-1025, doi: 10.1109/ACIT65614.2025.11185626.
- 12.** K. T. Selvi, S. S, V. H. Shetty, S. K. K and S. Shenoy, "Edutrack: AI based Personalized Learning Through Secure Analytics," *2025 4th International Conference on Innovative Mechanisms for Industry Applications (ICIMIA)*, Tirupur, India, 2025, pp. 1126-1131, doi: 10.1109/ICIMIA67127.2025.11200902.
- 13.** A. Banihashemi, A. Shahriary and Y. Yaghoobzadeh, "Human vs NotebookLM for Educational Podcasts: A Controlled Experiment on Two General Topics," *2025 15th International Conference on Computer and Knowledge Engineering (ICCCKE)*, Mashhad, Iran, Islamic Republic of, 2025, pp. 1-6, doi: 10.1109/ICCCKE68588.2025.11273844.

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Products and Systems

The landscape features diverse AI-driven tools emphasizing personalization, often integrating large language models (LLMs) for content generation and machine learning for adaptation. Google's NotebookLM processes uploaded sources to deliver tailored insights, explanations, and audio overviews, targeting researchers and students for free. Tutor AI creates custom courses on any topic, adapting paths for lifelong learners without specified pricing. Learn Your Way transforms PDFs into dynamic courses using AI-pedagogy blends, free with waitlist access. ChatGPT's Study Mode adjusts lessons based on user levels, offering quizzes and flashcards via subscription. Claude for Education provides Socratic guidance and templates for higher-ed users, with enterprise pricing. Khanmigo offers standards-aligned tutoring integrated with vast content, at \$4/month. StudyX generates notes, flashcards, and quizzes from uploads, with an Ultra plan for students. Gizmo focuses on spaced-repetition flashcards, "basically free." LearningStudioAI builds courses from documents with media integration, subscription-based. Diffit creates differentiated resources from inputs, free-tier for teachers. Magic School supplies 80+ tools for planning and assessments, with paid plans and security compliance. Brainly delivers adaptive quizzes and audio sessions, free with ads. Mindgrasp summarizes materials into notes and quizzes, subscription. MyWhiteboard.ai extracts info from videos and docs, pricing unspecified. Smart Sparrow enables adaptive lesson design for institutions, enterprise-priced.

Summary of Existing Research

AI-driven adaptive learning platforms and intelligent tutoring systems (ITS) significantly outperform traditional instruction by tailoring educational pathways to individual needs. Technologies like Deep Reinforcement Learning and NLP enable these systems to dynamically adjust content, fostering engagement and self-directed learning. Specific tools such as ChatGPT, Gemini, and NotebookLM are reshaping education, with studies showing AI-generated content—including educational podcasts—can rival human-created materials in quality and engagement. However, successful integration requires addressing data privacy, ethical concerns, and digital readiness. Ultimately, these AI innovations promise a more personalized, efficient, and accessible future for global education.

Gaps and Limitations Identified

Despite advancements, challenges persist. Privacy and security risks arise from data collection, with potential breaches. Algorithmic bias in training data can disadvantage marginalized groups, exacerbating inequities. Access barriers due to digital divides limit adoption in developing regions. Over-reliance on AI may erode critical thinking and emotional support from humans. Integration issues, technological costs, and lack of teacher training hinder implementation. Research gaps include few longitudinal studies, methodological inconsistencies, and underrepresentation beyond higher-ed or Western contexts. Ethical concerns like misinformation and lack of standardization call for frameworks.

Additionally, listed below are the platforms that inspired our work.

1. <https://notebooklm.google/>
2. <https://tutorai.me/>
3. <https://learnyourway.withgoogle.com/>
4. <https://gemini.google.com/gem/learning-coach>
5. <https://chatgpt.com/features/study-mode/>
6. <https://www.anthropic.com/news/introducing-claude-for-education>
7. <https://www.khanmigo.ai/>
8. <https://studyx.ai/>
9. <https://gizmo.ai/>
10. <https://learningstudioai.com/>
11. <https://app.diffit.me/#topic>
12. <https://www.magicschool.ai/>
13. <https://brainly.com/>
14. <https://app.mindgrasp.ai/>
15. <https://www.mywhiteboard.ai/>
16. <https://www.smartsparrow.com/>

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

7.1 Addressing Identified Research and Market Gaps

This research proposes the development of an **Integrated, Multimodal, and Language-Adaptive Learning Platform** to directly address the critical gaps identified in the current educational technology landscape. It moves beyond the prevailing model of fragmented, single-purpose tools by architecting a unified system where four specialized AI modules operate synergistically around a central learner model.

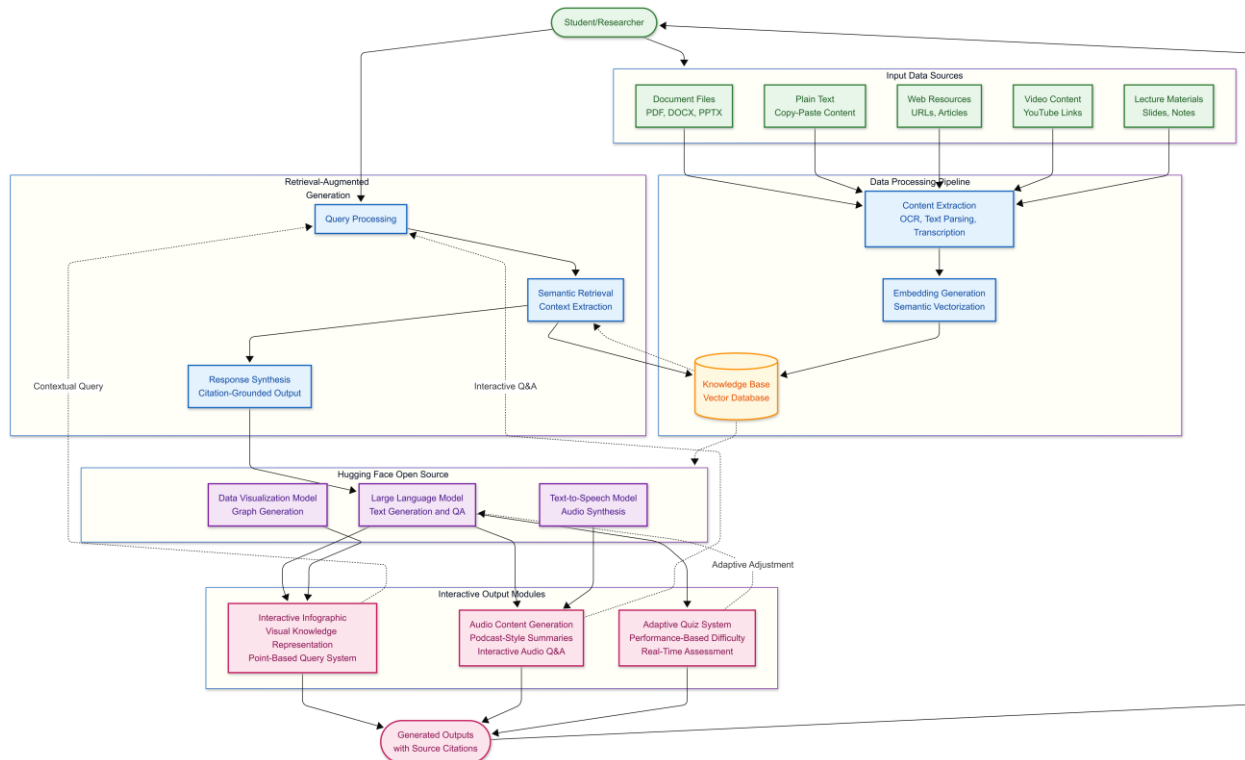
The solution specifically targets these gaps: (1) **The fragmentation between content creation, delivery, and assessment** by integrating multimodal synthesis, dynamic visualization, and adaptive testing into a single workflow; (2) **The lack of deep personalization in most tools** by employing a continuous knowledge-tracing model to drive real-time adaptation across all platform features; and (3) **The accessibility barrier for learners using informal mixed-language** by incorporating a dedicated natural language processing (NLP) engine for Sinhlish and Thanglish queries. This integrated approach ensures that personalization is not a superficial feature but the core, connective logic of the entire learning experience.

7.2 System Perspective: Input, Process, and Output

Input:

- **Source Materials:** Textual and multimedia inputs from educators/learners (PDFs, documents, web links, notes).
- **Learner Queries:** Questions and commands in English, Sinhlish, or Thanglish.
- **Implicit Feedback:** Behavioral data including interaction times, quiz responses, flashcard performance, and navigation patterns.
- **Pedagogical Parameters:** Learning objectives, desired difficulty, and content style preferences set by the creator or learner.

Conceptual Diagram



TAF-Conceptual Diagram.png

Process (Core AI Modules):

The system's intelligence is distributed across four interconnected processing modules

1. **Multimodal Media Factory:** Analyzes source text to automatically generate complementary learning assets, including scripted audio summaries, explanatory images/infographics, and composited video clips.
2. **Interactive Experience Tailor:** Adds an adaptive layer to generated media, embedding clickable elements and adjusting content complexity in real-time based on the evolving learner model.
3. **Adaptive Assessment Engine:** Utilizes a knowledge-tracing model to maintain a real-time estimate of learner proficiency, which it uses to generate personalized quizzes and provide formative, Socratic feedback.

These modules do not operate in isolation. They are coordinated by a **Central Knowledge Orchestrator**, which maintains a unified **Learner Model** and a **Knowledge Graph**. This graph links concepts, content assets, assessment items, and learner proficiency nodes, enabling consistent personalization across the entire platform.

Output:

- **Personalized Learning Interface:** A unified dashboard presenting the learner with (b) **interactive, personalized multimedia lessons**, (c) a **smart, spaced-repetition flashcard deck**, and (d) **adaptive quizzes** with targeted feedback.
- **System Intelligence:** An continuously enriched Knowledge Graph and refined Learner Model, which enhance the platform's adaptive accuracy for all users over time.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

1. How can members gain essential specialized domain expertise (AI, EdTech)?

- **Online Courses & Certifications:** We will enroll in specialized courses on platforms like Coursera, edX, and Udacity, focusing on "AI in Education," "Learning Analytics," and "Instructional Design" to build formal knowledge.
- **Research Papers & Journals:** We will regularly review the latest research papers from IEEE, ACM Digital Library, and Springer to understand state-of-the-art methods, such as using Reinforcement Learning for optimizing educational pathways.
- **Industry Resources:** We will follow research reports, blogs, and webinars from leading EdTech companies (e.g., Coursera, Khan Academy) to gain insights into current industry practices and challenges.
- **Supervisor & Research Group Guidance:** We will actively seek guidance and participate in discussions with professors and PhD scholars within our CoEAI (Centre of Excellence for AI) research group.

2. How can a member gain the technical knowledge and skills required for the project?

- **Core Programming (Python):** We will strengthen our Python programming skills, focusing on data structures and OOP concepts, through practice on platforms like LeetCode and HackerRank.
- **Machine Learning Frameworks:** We will complete tutorials and online courses (e.g., Andrew Ng's specialization) for libraries like Scikit-learn, TensorFlow, and PyTorch.
- **Natural Language Processing (NLP):** We will learn to use libraries such as NLTK, spaCy, and Hugging Face Transformers to build models for content analysis and interaction.
- **Practical Application:** We will develop small-scale projects (e.g., a simple recommendation system, quiz generator) to apply and test our technical skills practically.

3. How is the data required for the project collected?

Given the privacy constraints and scarcity of high-quality localized educational data, we will adopt a triangulated hybrid strategy to ensure system robustness:

- **LLM-Generated Synthetic Data (Primary Source for Content & Logic):** Since strict syllabus alignment is required, general datasets are insufficient. We will utilize Large Language Models (Gemini 3 Pro / Deepseek (Deepthink) via Knowledge Distillation to generate:
 - **Structured Course Content:** JSON-formatted modules derived from standard textbooks.
 - **Simulated Student Interactions:** Tanglish/Sinhlish chat logs and query variations to train the localized chatbot.
 - **Adaptive Quiz Scenarios:** Generating diverse question sets based on Bloom's Taxonomy.

Public Benchmarking Datasets (For Algorithms): To calibrate our "At-Risk Prediction" and "Knowledge Tracing" algorithms, we will utilize established educational datasets such as "EdNet" and "ASSISTments". This ensures our analytical models are grounded in proven statistical patterns of student behavior.

- **Controlled Pilot Study (Validation Stage):** Instead of collecting sensitive institutional data initially, we will conduct a closed-group pilot study with volunteer undergraduates. This will provide "Human-in-the-Loop" feedback on the generated content accuracy, UI usability, and the relevance of the AI's scheduling suggestions.
 - **Data Minimization:** Collecting only interaction logs necessary for performance tuning, without accessing sensitive personal records.
 - **Right to Withdraw:** Ensuring volunteers can opt-out at any time.

4. Does this project require ethical clearance?

Yes, ethical clearance will be required, specifically for the final validation phase. Since we are adopting a phased development approach, our ethical compliance strategy is two-fold:

Phase 1: Development (Synthetic Data - Exempt/Low Risk): During the initial development and training of the AI models, we will utilize Generated Synthetic Data and Public Datasets. As this involves no interaction with real human subjects or private institutional data, this phase poses minimal ethical risk. However, we will document our checks for Algorithmic Bias to ensure the model does not propagate unfair learning patterns.

Phase 2: Pilot Study (Human Participants - Clearance Required): For the final "Controlled Pilot Study" involving volunteer students, we will apply for ethical clearance from the Institutional Review Board (IRB). We will strictly adhere to protocols regarding:

- **Informed Consent:** Participants will be fully briefed on how the AI interacts with them.
- **Data Minimization:** Collecting only interaction logs necessary for performance tuning, without accessing sensitive personal records.
- **Right to Withdraw:** Ensuring volunteers can opt-out at any time.

9. Objectives and Novelty

To engineer a modular and interactive study suite using open-source models that outperforms the current capabilities of Google NotebookLM in the key areas of audio engagement, visual knowledge exploration, and personalized assessment.

Member Name with Registration No	Sub Objective	Tasks	Novelty
<i>Kojithan P.Y IT22264220</i>	To develop an intelligent, dynamic infographic generator that creates interactive and data-rich visual narratives from structured notes or research data.	. 1. Design a schema to extract and structure key entities, relationships, and statistics from user notes using an open-source LLM. 2. Implement multiple visualization templates (e.g., hierarchical, timeline, network) using open-source libraries 3. Develop an interface allowing users to dynamically filter data, click chart elements to drill down into details or source quotes, and toggle between visualization types. 4. Integrate a rule-based or lightweight AI model to suggest the most suitable chart type based on the analyzed data structure. 5. Enable real-time collaborative features, allowing multiple users to	Current tools, including NotebookLM, typically generate static or linear visual summaries. This component's novelty is its context-aware interactivity and user-driven narrative control . It transforms static charts into explorable data stories where users can interrogate the visualization (e.g., filtering, drilling down, switching perspectives) to uncover insights. Its use of open-source models for intelligent structuring and template suggestion, combined with real-time collaboration, creates a competitive, transparent alternative to proprietary systems.

		view, annotate, and discuss the same infographic simultaneously. 6. Create an export module to publish the visualization as an interactive HTML file or a static image for reports.	
<i>Baskaran V IT22172600</i>	To develop an interactive, AI-powered audio review component that transforms passive listening into an active learning and analysis session.	1. Implement automatic speech recognition using an open-source model to transcribe uploaded audio/podcasts. 2. Develop a processing pipeline that uses an open-source LLM to analyze transcripts for key concepts, generate structured summaries, and identify central themes. 3. Build an interactive web interface where users can click on any transcript segment to jump to that point in the audio and ask context-specific questions. 4. Integrate an open-source Text-to-Speech model to enable the system to verbally answer user queries and read summaries aloud. 6. Conduct system evaluation based on transcription accuracy, summary relevance, response latency, and user engagement metrics	Unlike Google NotebookLM's linear podcast output, this component introduces a bidirectional, conversational layer to audio content. The novelty lies in the integration of a cascade of specialized open-source models to create a queryable audio document . This allows for real-time, voice-based Q&A and assessment generation directly from the audio stream, promoting active comprehension. It directly competes by offering deeper interactivity and transparent, modifiable open-source architecture versus a proprietary black box.
<i>Nishara T IT22223876</i>	To build an Adaptive Quiz and Assessment Engine that personalizes	1. Develop a Knowledge Tracing Model to infer mastery of concepts from interaction history. 2. Implement an NLP pipeline to auto-generate diverse question types (MCQ,	The novelty is the tight integration of knowledge tracing with multi-format question generation. Quizzes are not a separate activity but a diagnostic probe that directly and visibly

	questions based on real-time learner performance and knowledge gaps.	fill-in, short answer) from source content. 3. Create an algorithm that adjusts quiz difficulty and topic focus based on the learner model. 4. Integrate a feedback mechanism that provides Socratic hints and explanations for incorrect answers, while also providing study resources via web scraping. 5. Evaluate the engine's impact on learning efficacy (pre/post-test scores).	updates the learner's personal knowledge model and mindmap, creating a closed learning loop.
<i>Sarmitha S</i> <i>IT22637482</i>	To develop an AI-powered handwritten notes digitization and multilingual study management system that converts handwritten note images into structured, translated, and study-ready materials and stores them in a searchable digital learning library.	1. Image Preprocessing: Apply resizing, binarization, noise removal, and edge detection to enhance handwritten images. 2. OCR Module: Use an open-source Optical Character Recognition model to extract handwritten text from images. 3. Multilingual Translation Module: Implement AI-based translation supporting Sinhala ↔ Tamil and Sinhala ↔ English to enable multilingual accessibility. 4. Notes Structuring Module: Clean extracted text, organize content into headings/topics, and highlight key concepts automatically. 5. Flashcard Generation Module: Automatically generate Q&A flashcards, summaries, and revision	Unlike existing OCR or note-scanning tools that only digitize handwritten content, this component introduces an end-to-end AI learning pipeline. The novelty lies in integrating handwriting extraction, multilingual translation, intelligent structuring, automatic flashcard creation, and digital library storage into one unified system. The platform transforms handwritten notes into searchable, translated, and reusable study resources, supporting multilingual students and improving long-term knowledge management.

		materials from structured notes. 6. Digital Library Storage: Store processed notes and flashcards in a searchable digital repository where users can retrieve materials using keywords or topics. 7. System Evaluation: Evaluate OCR accuracy, translation quality, retrieval efficiency, and usefulness of generated study materials.	
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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	
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- b) The proposed topic exhibits novelty.

Yes		No	
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- c) The student group can successfully execute the proposed project.

Yes		No	
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- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	
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- e) Any other comments:

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	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				